WEAK PRE-ADMISSIBILITY DESCENDS ALONG A TAUT MAP

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement opens with the chapter's phrase 'In the situation above', which fixes the two rings and the taut map; that situation is recorded as a standard prerequisite rather than delivered as a context span.

Editorial note. The counted spans carry no navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- The taut descent situation and faithful flatness of the map between the quotients. Same chapter, the situation recalled in the statement, used at proof lines 1832-1835. Allows a containment to be tested after passing to the quotients.
- Weak ideals of definition and topological nilpotence. Same chapter, the definitions in force in the statement. Fix what has to be shown for each element of the chosen open ideal.

Lemma 0.1. *In the situation above, if B is weakly pre-admissible, then A is weakly pre-admissible.*

Proof. Let $J \subset B$ be a weak ideal of definition. Let $I \subset A$ be an open ideal such that $IB \subset J$. To show that I is a weak ideal of definition we have to show that any $f \in I$ is topologically nilpotent. Let $I' \subset A$ be an open ideal. Denote $J' \subset B$ the closure of $I'B$. Then $A/I' \to B/J'$ is faithfully flat, hence injective. Thus in order to show that $f^n \in I'$ it suffices to show that $\varphi(f)^n \in J'$. This holds for $n \gg 0$ since $\varphi(f) \in J$, the ideal J is a weak ideal of definition of B, and J' is open in B. $\square$

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